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Supporting Information

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Second-Harmonic Generation with a 440 000% W^{-1} Conversion Efficiency in a Lithium Niobate Microcavity without Periodic Poling

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1. Etching depth of the microcavity

We simulated the transmission loss of $TE_{01,\,SH}$ at θ=0° considering the mode loss (**Figure S1**) using COMSOL software. It can be seen that when the etching depth is greater than 420 nm, the intrinsic quality factor Q_i is more significant than 10^{12}, which indicates the mode loss is much less than the absorption loss of lithium niobate material. Considering thin film lithium niobate is difficult to etch deeply, we chose the etch depth of 420 nm to perform experiments.

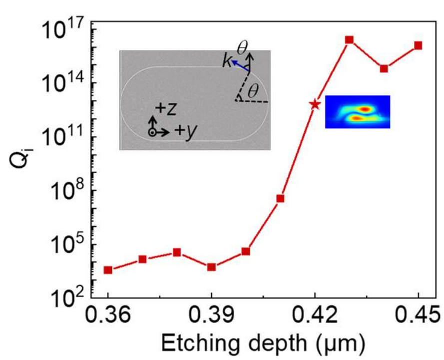

Figure S1. The simulated Q_i value of $TE_{01,\,SH}$, considering the losses due to mode coupling and radiation at θ=0.

2. Theory of Nonlinear Coupling Strength

The nonlinear coupling strength g is calculated according to [1-2]

$$g = \frac{d_{\mathrm{eff}}}{{n_{\mathrm{FH}}}^2 n_{\mathrm{SH}}} \sqrt{\frac{\hbar \omega_{\mathrm{FH}}^2 \omega_{\mathrm{SH}}}{2\varepsilon_0 L}} \zeta, \tag{S1}$$

where n_{FH} (n_{SH}) is the effective refractive index of the fundamental harmonic (FH) and second harmonic (SH), ω_{FH} (ω_{SH}) is angle frequency, and ε_0 is the vacuum dielectric constant. L is the perimeter of the resonator. d_{eff} is the effective nonlinear coefficient. ζ is the mode overlap. In the straight-arm region, ζ is denoted by

$$\zeta = \frac{\int_{\mathrm{LN}} d(x,z)\cdot E_{\mathrm{FH,z}}^{2} E_{\mathrm{SH,z}}^{*} \mathrm{d}x\mathrm{d}z}{\int_{all} \left|E_{\mathrm{FH}}\right|^{2} \mathrm{d}x\mathrm{d}z \sqrt{\int_{all} \left|E_{\mathrm{SH}}\right|^{2} \mathrm{d}x\mathrm{d}z}}, \tag{S2}$$

where $d(x,z)=\pm1$, represents the normalized nonlinear coefficient on the straight-arm. E_{FH} (E_{SH}) and $E_{\mathrm{FH,z}}$ ($E_{\mathrm{SH,z}}$) represent the electric field intensity and its z component for the FH and SH modes, respectively. In our waveguide with w_{ring}=1.27 μm, thickness of 600 nm, etching depth of 420 nm, ζ=0.83 μm^{-1}, g=6.35 MHz.

3. Quality Factor in the Visible Band

Since the SH wavelength is 759.2 nm and outside the wavelength range (765-781 nm) of the tunable laser in our lab, we cannot directly excite the $\mathrm{TE}_{01,\,\mathrm{SH}}$ mode in the microcavity in the main text with w_{ring}=1.27 μm. A typical transmission spectrum at around 770 nm is shown in **Figure S2**a. The microring can be coupled to more than five modes, as shown with arrows of different colors at specific polarization in some bands. The pink arrows give the position of the fundamental mode with a free spectral range (FSR) of ≈0.157 nm. The fundamental mode at 770.06 nm in the dashed line box is shown in Figure S2b with a loaded quality factor of 6×10^5.

We then measured the quality factor Q of the $\mathrm{TE}_{01,\,\mathrm{SH}}$ mode in a cavity on the same chip with w_{ring} of 1.21 μm. Through temperature tuning, the SH signal is strongest when the FH and SH are in double resonance. Therefore, we confirmed the position of the SH mode in the transmission spectrum, and the loaded Q of the SH mode was measured to be 1.4×10^4.

The quality factor $Q_{\mathrm{SH,01}}$ is two orders of magnitude smaller than the $\mathrm{TE}_{00,\mathrm{SH}}$. It is mainly due to the coupling of $\mathrm{TE}_{01,\,\mathrm{SH}}$ into other modes in the bending parts of the racetrack resonator. The TE refractive index in the X-cut cavity is varied periodically as

$$n_e(\theta) = \left(\frac{\cos^2\theta}{n_o^2} + \frac{\sin^2\theta}{n_e^2}\right)^{-1/2}, \tag{S3}$$

where n_{o}, n_{e} is the refractive of ordinary and extraordinary waves. According to the simulation, there are mode coupling in the curved part of the ring cavity for $\mathrm{TE}_{01,\mathrm{SH}}$ (**Figure S3**). Although the mode conversion process is adiabatic, the propagation loss of the converted mode is large, resulting in low $Q_{\mathrm{SH,01}}$. Fortunately, the mode in the microring will be $\mathrm{TE}_{01,\mathrm{SH}}$

at θ=±90° according to Equation S3, and the $TE_{01,SH}$ mode will always be involved in nonlinear processes in the straight waveguide.

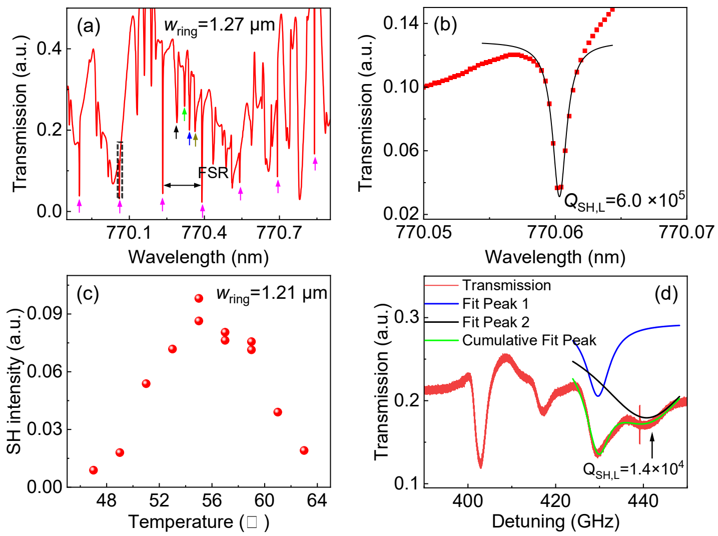

Figure S2. a) A typical transmission spectrum at 770 nm in a w_{ring}=1.27 μm microcavity. b) A zoomed fundamental-mode transmission spectrum with a loaded quality factor $Q_{00,SH}$=6.0×10^5. c) The variation of SH intensity with temperature in a w_{ring}=1.21 μm, which is on the same chip as the cavity investigated in the main text and was fabricated in the same batch microcavity. d) The $TE_{01,SH}$ mode transmission spectrum with a loaded quality factor $Q_{01,SH}$=1.4×10^4.

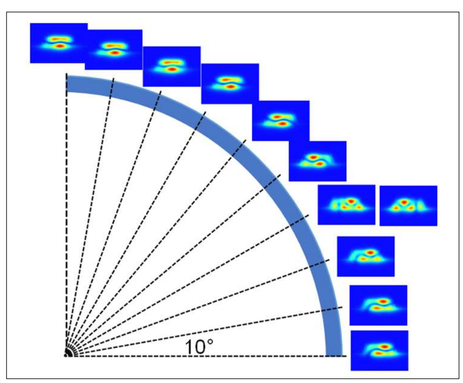

Figure S3. Simulated the eigenmode in the bent part of the race-tracked TFLN resonator.

4. Optical Parametric Oscillation Threshold

The output SH threshold power of the Optical Parametric Oscillation (OPO) process can be estimated using[1]

$$P_{\mathrm{SH}} \geq \hbar \omega_{\mathrm{SH}} \kappa_{\mathrm{SH,c}} \kappa_{\mathrm{s,m}} \kappa_{\mathrm{s,-m}} / 2g^2, \tag{S4}$$

where $\kappa_{s,m}$ and $\kappa_{s,-m}$ are the total dissipation rates of the OPO signals with the experiment values of $\kappa_{s,m}$=1.27 GHz, $\kappa_{s,-m}$=1.34 GHz. $\kappa_{SH,c}$ is the coupling dissipation rate of the SH, which can be deduced to be $\kappa_{SH,c}$=57.3 GHz according to $\kappa_c = \omega/2Q_c$. We calculated output SH power should be higher than 320 μW using Equation S4. According to the fitting curve of **Figure 3c** in the main text, the pumping threshold is 1.04 mW, which is close to the observed threshold of 1.21 mW.

The pump threshold power of the OPO process based on four-wave mixing can be estimated using[3]

$$P_{\mathrm{th}} \approx 1.54\left(\frac{\pi}{2}\right)\frac{Q_{\mathrm{FH,c}}}{2Q_{\mathrm{FH,L}}}\frac{{n_{\mathrm{neff}}}^{2}V}{n_2\lambda_{\mathrm{FH}}{Q_{\mathrm{FH,L}}}^{2}}, \tag{S5}$$

where n_2 is the nonlinear refractive index, $V=L\cdot A_{\mathrm{eff}}$ is the resonator mode volume. We substituted A=0.78 μm^2 and n_2=0.91×10^{-15} cm^2 W^{-1} into Equation S5, and then calculated the threshold of OPO as P_{th} = 350 mW, which is much higher than the threshold of cascaded OPO. Therefore, we attribute the OPO signal observed in the RP dual layer TFLN microcavities to the cascaded second-order nonlinear processes.

References

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